

Who Pays More, and When? Food Price Premiums and Abnormal Price Movements Across Selected Pakistani Urban Markets

Project Milestone 01 — Team Formation, Problem Framing & Data Curation | CS/SDP 312/314 L1, Fall 2026

1. Team

Yousuf Uyghur (Computer Science) and Sameer Hassan (Computer Science), with a third member joining the team. One member owns data acquisition, parsing and provenance control; the other owns statistical framing, representation and ethics analysis, and interpretation for a policy audience. Modelling is shared and reviewed by both.

2. Social Problem and SDG Connection

Food affordability in Pakistan is not experienced uniformly. National inflation figures describe an average, but households buy food in a specific city at a specific outlet, and the same essential item can cost materially different amounts in different urban markets in the same week. Where a city persistently pays above the cross-city median for staples, households there face systematically higher retail price exposure than the national headline reveals — a gap invisible in aggregate reporting, and one that may contribute to affordability pressure, though income and expenditure are not observed in this data.

This gives the project its framing. PBS publishes a single weekly SPI figure that is widely treated as the national food-price signal. That figure is a weighted index of average movement, and it is not designed to express geographic dispersion — but the dispersion is measurable from PBS’s own published city-level detail. We therefore ask what the headline necessarily omits: whether persistent city-level differences in food affordability sit underneath it, and whether the geographic detail published each week is sufficient to detect localised price stress.

Primary SDG: SDG 2 (Zero Hunger), specifically Target 2.1 on access to sufficient and affordable food and Target 2.c on food price monitoring and limiting extreme price volatility. **Secondary SDG:** SDG 10 (Reduced Inequalities), since the analysis concerns systematic differences in what comparable households pay across locations.

3. Dataset

Source. The Pakistan Bureau of Statistics (PBS) publishes a Sensitive Price Indicator (SPI) release every week. Each release includes an Annexure workbook whose Appendix sheets record, for each essential item and each city, the minimum, average, and maximum observed price together with the unit of measurement. PBS describes the SPI as covering 51 essential items collected from 50 markets across 17 cities, on base 2015–16 = 100.

How it is obtained. PBS does not publish a longitudinal dataset. It publishes hundreds of separate weekly workbooks, each on its own release page. We have written and tested a collector that discovers these pages, downloads each Annexure workbook, records a hash and retrieval timestamp for provenance, and parses the multi-level Appendix tables into tidy city–item–week records.

Curation performed. The curation contribution is longitudinal reconstruction and harmonisation across many separately published releases:

- Reconstructing tables with merged and multi-row headers, in which each city spans three MIN/AVG/MAX sub-columns, into one record per city–item–week.
- Normalising city labels that appear hyphenated across line breaks in the source (for example “Rawal-pindi”, “Gujran-wala”).
- Harmonising item labels and units, since items are quoted on different bases (a 20 kg flour bag against per-kilogram items) and label wording is not stable across releases.
- Handling explicit missing markers (“N.A.”) and blank cells as genuine missingness rather than as zero.
- Validating schema consistency release-to-release and deriving weekly percentage changes, cross-city median prices, city price premiums, and dispersion measures.

Scope. As of the current verified sample: 15 weekly releases retrieved and parsed between 2025-10-30 and 2026-08-27, yielding 5,460 city–item–week records across 7 cities and 52 item rows (16,380 individual price observations, since each record carries a minimum, average and maximum), with a 100% parse success rate on the verified set. The 52 parsed item rows sit against the 51 essential items PBS describes in its documentation; reconciling that difference — whether it reflects a split item, a subtotal row, or a basket revision — is part of the harmonisation work. The stable food commodities will be selected from these items. The verified weeks are not contiguous across that window, so extending the collector to close the gaps between releases, and to recover earlier weeks, is an active part of the curation work rather than an optional addition.

Limitations. First, the verified sample contains seven city groups — Islamabad, Rawalpindi, Gujranwala, Sialkot, Lahore, Faisalabad and Sargodha — against the 17 cities PBS describes, and a second Appendix sheet that may carry the remaining cities is still being examined. Until that is resolved, the analysis is described as covering selected urban markets rather than the country as a whole. Second, because the verified weeks are not yet contiguous, week-over-week change and lagged features are computable only where consecutive releases exist, so usable observations for any lag-based analysis are fewer than the total record count. Third, SPI records retail prices only, so we observe the downstream outcome of food distribution and not upstream causes such as harvests, transport or inventories; we report associations, not supply-chain causation. Fourth, the MIN–MAX range reflects variation within PBS’s sampled markets and outlets and may also capture quality, outlet, or locality differences, so it will not be interpreted as a pure measure of market fragmentation. Fifth, comparability of item definitions before the current base period is unestablished, so the modelling window is restricted to releases verified to share a consistent structure.

Sources considered and rejected. We evaluated merging PBS retail prices with Punjab AMIS wholesale mandi prices to study wholesale-to-retail transmission. Testing showed AMIS serves current prices but exposes no reproducible historical query, so the merge was abandoned before it could consume project time — a decision that shaped the final scope.

4. Initial Hypotheses

- **H1 (persistent premiums).** Differences in essential-food prices across cities are persistent rather than transient: cities sitting above the cross-city median in one week tend to remain above it in following weeks.
- **H2 (perishability and dispersion).** Perishable commodities such as tomatoes, onions and potatoes exhibit greater cross-city price dispersion than storable staples such as rice, pulses and sugar. The related claim that perishables are also more volatile week to week is separated deliberately: dispersion is a within-week quantity and is measurable on the sample as it stands, whereas volatility requires consecutive releases and is therefore contingent on the backfill.
- **H3 (predictability of abnormal movement).** Recent price behaviour carries predictive information about subsequent abnormal price movements, and the direction of that relationship may differ between perishable and storable commodities — momentum in one, mean reversion in the other, or no relationship at all. “Abnormal” is defined relative to each commodity’s own historical distribution rather than as a fixed percentage.
- **H4 (representation and model performance).** The city-level detail published alongside the weekly headline is uneven in geographic reach, and predictive performance may vary across locations. We will quantify the coverage actually obtainable from the releases and test whether error rates and abnormal-movement detection differ systematically by city.

Cross-city dispersion, persistence and coverage (H1, the dispersion half of H2, and H4) are the primary analysis because they are answerable with the data already verified. Next-week classification (H3) is secondary and will be scaled to the number of consecutive weekly releases we recover.

Methods. H1 separates two quantities. Average city premiums are estimated by regressing log price on city, item and week indicators under sum-to-zero coding, so each coefficient is a premium relative to the average city effect; persistence is tested directly, as the probability that a city above the cross-city median in one week remains above it in the next, with consecutive-week rank correlations. Because the first is mean-based and the second median-based, they are reported as distinct estimates. H2 compares cross-city dispersion between perishable and storable groups, with the volatility comparison and the clustering of city–item series into a market typology run once consecutive releases permit it. H3 treats abnormal next-week movement as binary classification, logistic regression as baseline and ensemble methods as comparison, under a time-ordered split rather than random folds; lagged features exist only within consecutive-release pairs, so the usable sample for H3 is a subset of the panel and will be reported as such. H4 is addressed descriptively and, in the fairness milestone, by testing whether classifier performance differs systematically across cities, stratified by population from the PBS census since SPI carries no measure of market size.

5. Future Extension (Week 14)

PBS publishes a new SPI release every week. We will freeze the training panel at a cutoff in late September 2026, chosen early so that the holdout accumulates as many genuinely new waves as possible before Week 14, and record every subsequent weekly release into a separate holdout store that the collector appends to but never edits. At Week 14 we will first evaluate the frozen model on observations collected after the original cutoff. We will then incorporate an earlier portion of the newly available data into the training set and evaluate the retrained model on a later, still-unseen portion, allowing a fair comparison of model stability and of the effect of retraining without evaluating any model on data it was trained on. We expect degradation on the post-cutoff weeks, concentrated in perishables whose behaviour is most seasonal, with retraining recovering much but not all of it; persistent city premiums we expect to hold across the cutoff. If required, a second extension adds historical weeks recovered by the backfill.